

Zoology Paper-I

VCD: 26/10/23

Hour: 03 Hours

Marks: 100

Please check whether you have got the right question paper.

N.B. 1. All questions are compulsory.

2. Figures to the right indicates full marks.

3. Draw neat and labelled diagrams wherever necessary.

Q. 1. A) Fill in the blanks.

5 Marks

- _____ is a nucleoside containing pyrimidine base.
(Adenosine, Guanosine, Cytidine, Inosine)
- _____ is the location of a particular gene on the chromosome.
(Locus, Genome, DNA, Allele)
- Humans have _____ pairs of autosomes and one pair of sex chromosome.
(44, 23, 22, 46)
- _____ is a Greek word which means stopping or suppression of gene.
(Epistasis, Pedigree, Codominance, Pleiotropy)
- _____ form of DNA has left-handed double helix.
(A-DNA, B-DNA, Z-DNA, H-DNA)

B) Match the column.

5 Marks

- | | |
|-------------------------------|--|
| 1. Metacentric chromosome | a) B- DNA |
| 2. AUG | b) 9:7 |
| 3. Affected male | c) Initiation codon |
| 4. Double recessive epistasis | d) V- shaped |
| 5. Rosalind and Franklin | e)  |

C) True or False.

5 marks

- DNA contains ribose sugar.
- Enzyme helicase help in unwinding of DNA strand.
- Colour-blindness is an example of X linked recessive inheritance.
- Type AB is a Universal donor blood group.
- Roan coat colour in cattle in an example of codominance.

D) Answer in one sentence.

5 Marks

- Define cistron.
- What is the phenotypic ratio of Mendelian Dihybrid cross?
- What is the shape of Acrocentric chromosome?
- What is heterochromatin?
- Which enzyme carries out proof reading of newly formed DNA?

20 Marks

Q. 2. Answer the following (Any 2)

1. What is Epistasis? Give a detailed account on Dominant Epistasis.
2. Describe Co-dominance with suitable example.
3. Describe in detail dihybrid cross and state the Mendelian principles of inheritance derived from it.
4. Describe the pattern of inheritance of blood group and Rh factor in Man.

20 Marks

Q. 3. Answer the following (Any 2)

1. Classify chromosomes on the basis of position of centromere.
2. Explain role of environment and hormone in sex determination.
3. Explain any two mechanisms of chromosomal basis of sex determination.
4. Explain the inheritance of colour blindness in Man.

20 Marks

Q. 4. Answer the following (Any 2)

1. Describe Griffiths transformation experiment.
2. Describe process of DNA replication.
3. Write a detailed account on types of DNA.
4. Give an account of Hershey Chase experiment of bacteriophage infection.

20 Marks

Q. 5. Write short note. (Any 4)

1. Structure of t-RNA.
2. One gene one enzyme hypothesis.
3. Polytene chromosome
4. Euchromatin and heterochromatin
5. Parthenogenesis
6. Lethal genes
7. Differentiate between test cross and back cross.
8. Characteristics of Autosomal dominant traits.

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